

Preliminary Study on Effect of Secretly Increasing or Decreasing Predicted Number of Steps to Promote Walking

Masami Takahashi

NTT Social Informatics Laboratories, NTT Corporation
Yokosuka-shi, Kanagawa-ken, Japan
msm.takahashi@ntt.com

ABSTRACT

With the widespread use of wearable devices, it is becoming easier to predict the future of an individual's behavior regarding health. However, little is known about how presenting future predictions about an individual's behavior affects awareness and behavior. The "StepUp Forecast" project is investigating walking behavior to clarify the impact of presenting predictions of the number of steps an individual will take on human behavior. In a previous study, it was found that self-efficacy and the number of steps taken increased significantly when the predicted number of steps was presented, compared with when only a record of the number of steps taken was shown. To advance this research, we investigated the effects of intentionally and secretly increasing or decreasing the predicted value rather than simply presenting it. Specifically, we observed changes in awareness and behavior when users were presented with a predicted value with 1,000 steps added or subtracted. In our 5-week experiment, 40 participants used StepUp Forecast, an application that presents the predicted number of steps on the basis of past life logs. The results indicate a non-significant but increasing trend in self-efficacy and number of steps added or subtracted compared with when only the step-count record was presented.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Ubiquitous and mobile computing design and evaluation methods**; • **Applied computing** → **Health care information systems**.

KEYWORDS

mHealth, Personal Informatics, Physical Activity, Self-Efficacy

ACM Reference Format:

Masami Takahashi. 2023. Preliminary Study on Effect of Secretly Increasing or Decreasing Predicted Number of Steps to Promote Walking. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*, October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3594739.3610683>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0200-6/23/10.

<https://doi.org/10.1145/3594739.3610683>



Figure 1: Outline of StepUp Forecast

1 INTRODUCTION

Information on future predictions, such as weather forecasts and traffic-jam predictions, has become a familiar part of our daily lives. People will sometimes make decisions about their daily activities on the basis of predictive information. However, it is less clear how presenting future predictions about personal behaviors regarding health affects attitudes and behaviors. For example, weather forecasting does not enable individuals to control the weather, so they make decisions about how they will act in the predicted future. If we consider, for example, the prediction of an individual's total daily steps, it is possible to change the predicted future by the individual consciously changing their behavior so that they walk more than usual.

Through our "StepUp Forecast" project [1], we are engaged in research to determine the impact of presenting personal-step-count forecasts on human behavior. We previously developed the application "StepUp Forecast" (see Fig. 1), which predicts and presents personalized step counts on the basis of past life logs, and experimentally tested the impact of presenting step-count predictions on awareness and behavior. In that previous study, we aimed to clarify the effects of presenting the predicted number of steps on self-efficacy and the promotion of walking behavior. In the experiment we conducted for that study, participants were divided into two groups. The normal prediction group was presented with the predicted number of steps, and the added-prediction group was presented with the predicted number of steps with an "explicit" addition of 1,000 steps. A screenshot of the prediction-presentation period from this experiment is shown in Fig. 2. Participants in the added-prediction group were explicitly told that the predicted value would be augmented by an additional 1,000 steps per day the additional steps would be displayed in a different color. In the experiment, changes in the number of steps and self-efficacy were compared between the monitoring period in which only the actual number of steps was presented and the prediction period in which the predicted number of steps was presented. The results indicated that both the number of steps and self-efficacy increased significantly in the normal prediction group. However, neither steps nor self-efficacy increased significantly in the added-prediction group. This indicates that when presenting a predicted number of steps,

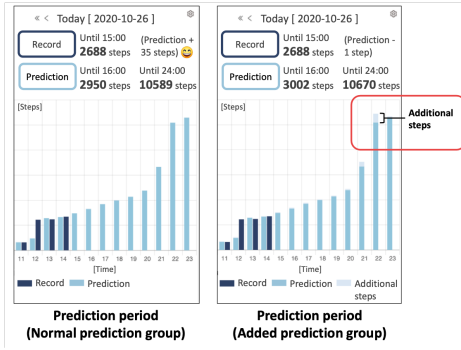


Figure 2: Previous screenshots of StepUp Forecast. Normal prediction and added-prediction groups during prediction period.

exceeding the prediction should be achievable for the user and that a positive loop that promotes behavior may occur by increasing self-efficacy through the experience of achievement beyond the prediction value.

One point that we were unable to clarify in the previous experiment, and the question we get asked the most, is “What would happen if we were secretly adding to our step count?”. This is the first research question of this study. The second research question is that if, as in the normal prediction group, the experience of walking beyond the predicted number of steps increased self-efficacy and promoted walking, then intentionally presenting a lower prediction might increase the experience of walking beyond the predicted number of steps and further promote walking. However, exceeding low predictions may be too easy and reduce motivation, but this point is also not clear. Therefore, this paper aims to answer the following research questions:

RQ1: Does secretly “adding to” the predicted the number of steps affect self-efficacy and promotion of walking behavior?

RQ2: Does secretly presenting a reduced number of predicted steps affect self-efficacy and the promotion of walking behavior?

2 RELATED WORKS

According to Bandura’s Social Cognitive Theory [2], a person taking a certain action is influenced by both outcome expectations and self-efficacy (Fig. 3). Therefore, we believe that certain behaviors may be prompted if we can present predictions that increase a person’s self-efficacy and outcome expectancy. In Bandura’s Social Cognitive Theory, behavior refers to, for example, walking or sleeping, and outcome refers to the result of the behavior, for example, a good mood or healthy state. Self-efficacy is the conviction that one can successfully execute the behavior required to produce the outcomes, and outcome expectation is defined as a person’s estimate that a given behavior will lead to certain outcomes. Previous studies have used life logs to predict behavior and outcome, which corresponds to the behavior prediction and outcome prediction that we added to the bottom half of Fig. 3.

Several studies focused on making predictions about individual attitudes and behaviors, examined the effects of presenting “outcome predictions”, such as an individual’s mood [3], postprandial

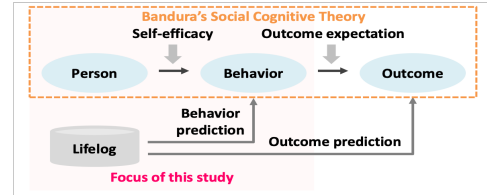


Figure 3: Diagrammatic representation of Bandura’s Social Cognitive Theory (top half) and prediction of behavior and outcome using life logs (bottom half).

blood glucose [4][5], weight [6], and stress events [7], and confirmed improvement in exercise and eating behaviors as a result of increased outcome expectations. “Behavior predictions” are specific and easy to imagine what the user should do compared with outcome predictions; however, there have been few studies on the impact of presenting behavior predictions. Takeuchi et al. [8] examined the effect of presenting behavior predictions by developing an application for forecasting future spending behavior, such as “You will spend 400 yen around Yayoi, Bunkyo, Tokyo after one hour”. However, they asked participants if their consumption amounts had changed but did not investigate whether those amounts had increased or decreased, so the impact on self-efficacy and behavior has yet to be examined in detail. Therefore, it is not clear what impact the presentation of behavior predictions has on self-efficacy and certain behaviors, especially in promoting desirable behaviors such as those regarding health.

3 STEPUP FORECAST

StepUp Forecast is a web application running on the Chrome browser of the Android platform that predicts the number of steps on the basis of past life logs and presents them to the user. The input data are shown in Table 1. The predicted number of steps is calculated using not only the user’s past steps but also weather, mood, and schedule. Note that the weather data are from the OpenWeatherMap API¹, step count and other data are from the Fitbit API², and daily mood and schedule data are manually input by the user. The output is visualized as an hourly cumulative graph, as shown in Fig. 1.

We constructed a model to predict the number of steps for each participant using Gradient Boosting from the past two weeks of data. Gradient Boosting is a machine learning algorithm that uses an ensemble of weak learners to minimize the impact of redundant explanatory variables. The model’s construction is based on an ensemble of weak learners so that even if the explanatory variables are redundant, they have little impact. Since we did not know in advance which explanatory variables would be valid for each participant, we chose Gradient Boosting to reduce the impact of redundant explanatory variables during model construction. We used `sklearn.ensemble.GradientBoostingRegressor` in the `scikit-learn` package [9] to set the hyperparameters of the model so that the mean absolute error of the training data would be less than 1,000 (see our previous study Chapter 3 for more details [1]).

In this experiment, users were presented with values that increased or decreased by a total of 1,000 steps per day to test the

¹<https://openweathermap.org/>

²<https://dev.fitbit.com/>

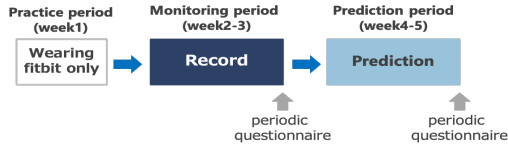


Figure 4: Schedule and conditions during experiment

effect of increasing or decreasing the predicted values on their awareness and behavior. Since the graphs were visualized on an hourly basis, the value was allocated according to the predicted number of steps per hour so that the total increase or decrease would be 1,000 steps in one day. This value of 1,000 steps was set on the basis of the government’s National Health Promotion Declaration [10], considering it a reasonable range to be added.

4 EXPERIMENT

The purpose of this experiment was to clarify the effects of intentionally and secretly increasing or decreasing the number of steps and presenting them to the user on self-efficacy and walking promotion. Through a monitoring recruitment company, 40 participants (21 men and 19 women) who were not receiving lifestyle-related treatment or guidance were recruited. Eight participants were in their 20s, 10 in their 30s, 10 in their 40s, and 12 in their 50s. Approval from our company’s Institutional Review Board was obtained.

4.1 Conditions

Participants wore the Fitbit only for the first week (practice period) then wore the Fitbit and used the StepUp Forecast application for the next four weeks (Fig. 4). The reason for the initial Fitbit-only period was to familiarize the participants with the operation and charging of the Fitbit, as some were using it for the first time. The first two weeks of the four-week period in which the StepUp Forecast application was used was a period in which only step-count records were presented (monitoring period), and the second two weeks were a period in which both step-count records and forecasts were presented (prediction period). During the latter two weeks, the group was divided into two groups: one group was presented with 1,000 steps added to the predicted number of steps, and the other group was presented with 1,000 steps subtracted. In allocating the groups, t-tests were conducted to confirm that there was no bias in the step counts between the groups by referring to the step-count records of the monitoring period.

The screen during the monitoring period is shown in Fig. 5(a) and that during the prediction period is shown in Fig. 5(b). The display for the monitoring period (Fig. 5(a)) showed the date and total number of steps from the last update at the top, and a bar graph at the bottom showing the total number of steps recorded per hour. The display during the prediction period (Fig. 5(b)) showed the date and total number of steps from the last update at the top, predicted number of steps until the next update, and total predicted number of steps for that day. The increase or decrease in the predicted number of steps was calculated behind the scenes depending on the group, but the user interface (UI) is displayed in such a way that the intentional increase or decrease did not appear to the user.

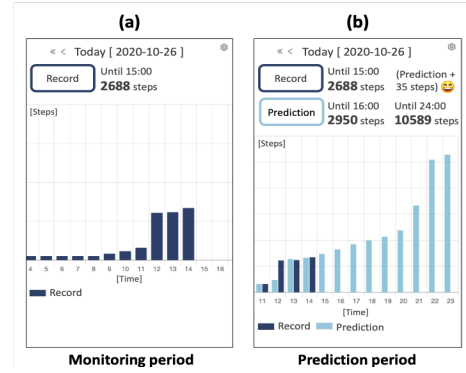


Figure 5: Screenshots of StepUp Forecast. (a) During monitoring period, (b) during prediction period.

4.2 Procedure

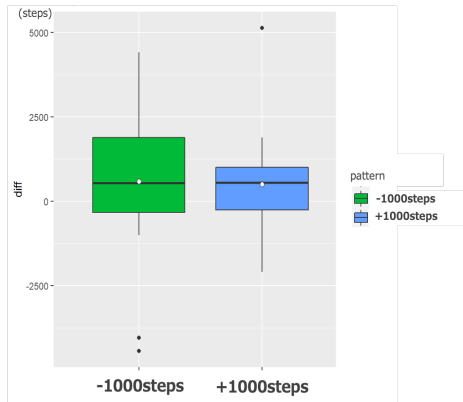
The experiment was conducted over a five-week period from mid-January to late February 2022. Prior to the experiment, participants were informed that this was an evaluation experiment of an application targeting walking and that they were expected to use and evaluate the application in their normal daily activities. Because it was winter and there was concern about the re-emergence of COVID-19, we did not give any instructions to forcibly promote walking for duration of the experiment. Participants were given a Fitbit versa2 and instructed to wear it at all times, except when bathing and charging. We also explained that during the prediction period, the predicted number of steps would be calculated from data collected in the past using Fitbit and other devices and visualized as a bar graph. Information on daily mood and schedule, which would be the data used for the prediction model, was obtained by having the participants respond to a questionnaire displayed when the application was first launched each day. During the experiment, a prediction model was constructed for each participant and updated daily.

4.3 Data Collection

To examine differences in step counts and self-efficacy when predictions were displayed compared with when only step-count records were presented, each participant’s total daily steps were collected via Fitbit and self-efficacy values were collected via periodic questionnaires. Questionnaire to assess self-efficacy was developed on the basis of a previous study [11], after completing 1,000, 2,000, and 20,000 steps, participants were asked to indicate on a scale of 0 (not at all confident) to 100 (completely confident) how confident they were in their daily walking ability. 1,000 steps, 2,000 steps,... and 20,000 steps, defining self-efficacy as the average value of confidence for each of the 20,000 steps. In the daily questionnaire to obtain mood and schedule data for use in the predictive model, we also asked whether there were any special circumstances, such as being significantly ill and sleeping during the day before, to determine whether the previous day’s data should be treated as outliers. After the experiment, all participants were interviewed about their impressions of the application during each period, whether it had affected their motivation and behavior to walk, and the impact of intentionally increasing or decreasing the number of steps by 1,000.

Table 1: Explanatory variables for prediction model

Feature name	Description
Day of the week	from Monday to Sunday
Weather	clear, cloudy, rain
Schedule	work at office, business trip, work at home, holiday (going out), holiday (at home), trip
Mood	Five-point Likert scale from very bad (1) to very good (5)
Steps (one week ago)	Total number of steps walked one week ago
Steps (last hour)	Number of steps walked in last hour
Steps (last 15 minutes)	Number of steps walked in last 15 minutes
Steps (same time one week ago)	Number of steps walked in same amount of time one week ago
Cumulative steps	Cumulative number of steps
Heart rate	Average heart rate in last 15 minutes

**Figure 6: Average difference in number of steps over monitoring and prediction periods (box plot)**

Participants were told during the interview whether they belonged to the group that increased or decreased the number of predicted steps by 1,000.

5 RESULTS

Figure 6 is a box-and-whisker plot showing the difference in the average number of steps taken during the monitoring and prediction periods. The Shapiro-Wilk test was conducted to check for normality in the differences in the number of steps between the monitoring and forecast periods. Since some of the differences were not normal, multiple comparisons (using Holm) were conducted after the Mann-Whitney U test with p-values adjusted. Regarding the difference in the number of steps between the monitoring and prediction periods, it was found that although the number of steps tended to increase during the prediction period, there was no significant difference at the 5% level between the -1000-step group and +1000-step group. The self-efficacy data for the monitoring and forecasting periods were also analyzed, and it was found that, similar to the difference in the number of steps, there was an increasing trend, but there was no significant difference at the 5% level between the -1,000-step and +1,000-step groups.

6 DISCUSSION AND LIMITATIONS

Regarding RQ1, when +1000 steps were secretly added to the predicted number of steps, the number of steps increased during the

prediction period, although not significantly. It is possible that the number of steps did not increase in the previous experiment due to the effect of the UI clearly indicating the amount of the addition. Regarding RQ2, when the predicted number of steps was secretly reduced by -1000 steps, there was a non-significant but increasing trend in the number of steps during the prediction period. Although it depended on the degree of reduction, -1000 steps may increase the experience of walking more than predicted and promote the number of steps taken. The extent to which it is appropriate to reduce the number of steps is an issue for future study. From the results of this study, it was found that when presenting prediction to promote healthy behavior, exceeding the predicted value should still be achievable for the user. Around -1000 steps would not be counterproductive even if presented in a reduced form, and through the achievement experience it is believed that positive loops may be generated by increasing self-efficacy. It was also suggested that when presenting an additional step, not explicitly showing that there is an additional step may promote [healthy | walking?] behavior.

Since the experiment was conducted during the COVID-19 pandemic, when people refrained from going out unnecessarily, the results may change if people can go out more frequently. Furthermore, since the monitoring and prediction periods were only two weeks each, and the number of participants was not large (20 per group), further experiments are needed to verify longer-term and more reliable effects.

7 CONCLUSIONS

We investigated the impact of secretly adding or subtracting the number of steps predicted when using the application “StepUp Forecast”, which presents step prediction on the basis of past life logs. The results indicate a trend toward an increase in the number of steps and self-efficacy, although the difference was not significant. The results also indicate that when prediction is presented to promote healthy behavior, exceeding the predicted value should still be achievable for the user and that presenting a reduced value would not be counterproductive and could create a positive loop by increasing self-efficacy through the experience of achievement. The results suggest that when the number of steps is increased, not explicitly indicating the extent to which it was increased may promote healthy behavior. The experiment for this study was conducted when COVID-19 re-emergence occurred, and some participants

stated during interviews that they had tried to refrain from unnecessary outings. We would like to conduct the experiment again in a situation without these restrictions.

REFERENCES

- [1] Asuka Miyake, Masami Takahashi, Ryo Hashimoto, and Momoko Nakatani. 2021. StepUp Forecast: Predicting Future to Promote Walking. In Proceedings of the 23rd International Conference on Mobile Human-Computer Interaction (MobileHCI '21). Association for Computing Machinery, New York, NY, USA, Article 16, 1–12.
- [2] Albert Bandura. 1977. Self-efficacy: Toward a Unifying Theory of Behavioral Change. *Psychological review*. Vol. 84, No. 2, 191–215.
- [3] Victoria Hollis, Artie Konrad, Aaron Springer, Matthew Antoun, Christopher Antoun, Rob Martin, and Steve Whittaker. 2017. What Does All This Data Mean for My Future Mood? Actionable Analytics and Targeted Reflection for Emotional Well-Being. *Human-Computer Interaction* 32, 5–6 (2017), 208–267.
- [4] Pooja M. Desai, Elliot G. Mitchell, Maria L. Hwang, Matthew E. Levine, David J. Albers, and Lena Mamykina. 2019. Personal health oracle: Explorations of personalized predictions in diabetes self-management. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. ACM, 1–13.
- [5] Bryan Gibson, Leah Yingling, Alisa Bednarchuk, Ashwini Janamatti, Ingrid Oakley-Girvan, and Nancy Allen. 2018. An Interactive Simulation to Change Outcome Expectancies and Intentions in Adults With Type 2 Diabetes: Within-Subjects Experiment. *JMIR Diabetes* 3, 1 (2018), e2.
- [6] Saeyoung Rho, Injung Lee, Hankyung Kim, Jonghyuk Jung, Hyungi Kim, Bong Gwan Jun, and Youn-kyung Lim. 2017. FutureSelf: What Happens When We Forecast Self-Trackers? Future Health Statuses? In Proceedings of the 2017 Conference on Designing Interactive Systems (DIS '17). ACM, 637–648.
- [7] Kwangyoung Lee, Hyewon Cho, Kobiljon Toshnazarov, Nematjon Narziev, So Young Rhim, Kyungsik Han, YoungTae Noh, and Hwajung Hong. 2020. Toward Future-Centric Personal Informatics: Expecting Stressful Events and Preparing Personalized Interventions in Stress Management. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20). ACM, 1–13.
- [8] Toshiki Takeuchi, Takuji Narumi, Kunihiro Nishimura, Tomohiro Tanikawa, and Michitaka Hirose. 2010. Receiptlog Applied to Forecast of Personal Consumption. In 2010 16th International Conference on Virtual Systems and Multimedia. IEEE, 79–83.
- [9] scikit-learn. `sklearn.ensemble.GradientBoostingRegressor`. Retrieved from <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.GradientBoostingRegressor.html>.
- [10] National Health Promotion Movement in the twenty first century. Ministry of Health, Labour Standards. Retrieved June 2, 2023 from <http://www.kenkounippon21.gr.jp/kenkounippon21/about/kakuron/index.html>
- [11] Albert Bandura and Nancy E. Adams. 1977. Analysis of Self-Efficacy Theory of Behavioral Change. *Cognitive therapy and research*, 1, 4, 287–310. <https://doi.org/10.1007/BF01663995>